Problem Statement: A Worldwide EDA Tool Dataset, Built on Crucible

Background

Crucible is Ignytion's reproducible tooling layer for hardware and silicon design. Its command-line interface, `igny`, installs tools and Python libraries, keeps them isolated in a named environment, and locks the result so another person can reproduce it exactly.

Information about EDA tools is scattered across vendor websites, open-source repositories, documentation, and community lists. There is no single structured place to see which tools exist worldwide, who makes them, and what they do.

This project builds that dataset, and it uses Crucible as the platform the whole project runs on.

Problem

Build a complete data science project that runs entirely on Crucible and collects information about EDA tools from all over the world.

The subject is the global EDA landscape: commercial and open-source tools from every vendor, community, and region. It is not limited to the tools Crucible can install. Crucible is where the project runs, not what the project studies.

"Entirely on Crucible" means every tool, Python library, and script the project uses is installed and run through `igny`. Nothing is installed with a system-wide `pip install`, a system package manager, or a manual `PATH` edit.

Objectives

1. Set up the project as a Crucible workspace bound to its own environment.
2. Install the data science stack through `igny`.
3. Identify public sources of information about EDA tools worldwide.
4. Write collection scripts that gather the information from those sources and run them through `igny`.
5. Clean the collected information into one structured dataset, with duplicates merged.
6. Analyse the dataset and present the findings with plots.
7. Lock the workspace so the whole project can be reproduced on another machine.

Data to collect

One row per EDA tool. The fields below are a minimum; add more where they are useful.

Field	Example
Tool name	Design Compiler, Yosys
Vendor or maintainer	Synopsys, YosysHQ
Country or region of origin	United States, Austria
Category	synthesis, simulation, place and route, verification, layout, analog, PCB
Licence model	commercial, open-source, freeware
Licence	proprietary, ISC, GPL
Website or repository URL	the canonical public link
First release year	when the tool first shipped
Latest version and release date	the most recent public release
Supported platforms	Linux, Windows, macOS
Supported languages and formats	Verilog, VHDL, SystemVerilog, GDSII
Source of the record	the URL the row was collected from
Date collected	when the row was collected

Coverage requirements:

- Both commercial and open-source tools.
- Every major category in the EDA flow, from front-end design to physical sign-off.
- Vendors and communities across North America, Europe, and Asia at a minimum.

Typical sources are vendor product pages, open-source repositories and their public APIs, public documentation, and published lists of EDA software.

Constraints

- Every command that runs a tool or a script goes through `igny run` or `igny run script`, or runs inside `igny env shell`.
- Every dependency is installed with `igny tool install` or `igny suite install`.
- If a library the project needs is not in the Crucible catalog, do not install it another way. Use what the catalog and the Python standard library provide, and record the missing library as a gap in the final report.
- Collect public information only. Follow each source's terms of use and `robots.txt`, and keep request rates low.

- Every row records where it came from and when. No hand-entered rows without a source, and no made-up values. A field that cannot be found stays empty.
- The workspace is locked with `igny workspace lock`, and `crucible.toml` and `crucible.lock` are committed together.

Deliverables

1. A Crucible workspace containing `crucible.toml` and `crucible.lock`.
2. Collection scripts, one per source or source type.
3. A dated snapshot of the raw collected data.
4. The cleaned dataset in CSV format.
5. An analysis notebook or script with plots, for example tools per category, commercial against open-source, tools per region, and release activity over time.
6. A short report covering the sources used, what the data shows, known gaps in coverage, and every place where Crucible fell short for data science work.
7. A README with the exact steps to reproduce the project.

Acceptance criteria

The project is complete when a person who has never seen it can:

1. Clone the repository and run `igny workspace sync`.
2. Run the collection scripts with `igny run script` and get a fresh dataset from the live sources.
3. Run the analysis on the committed snapshot and get the same plots as the report.
4. Trace any row in the dataset back to its source URL.
5. Do all of the above without installing anything outside Crucible.

Getting started

```
igny env create eda-data
igny workspace create eda-data-science --path ./eda-data-science --env eda-data
cd eda-data-science

igny catalog --view full          # see which libraries Crucible can provide

igny tool install numpy
igny tool install matplotlib
igny tool install jupyterlab

igny run script ./scripts/collect.py --log
igny workspace lock
```

The catalog changes between releases. Check `igny catalog` on the installed CLI before relying on any library named here.